

1.3 Derivatives for Optimization

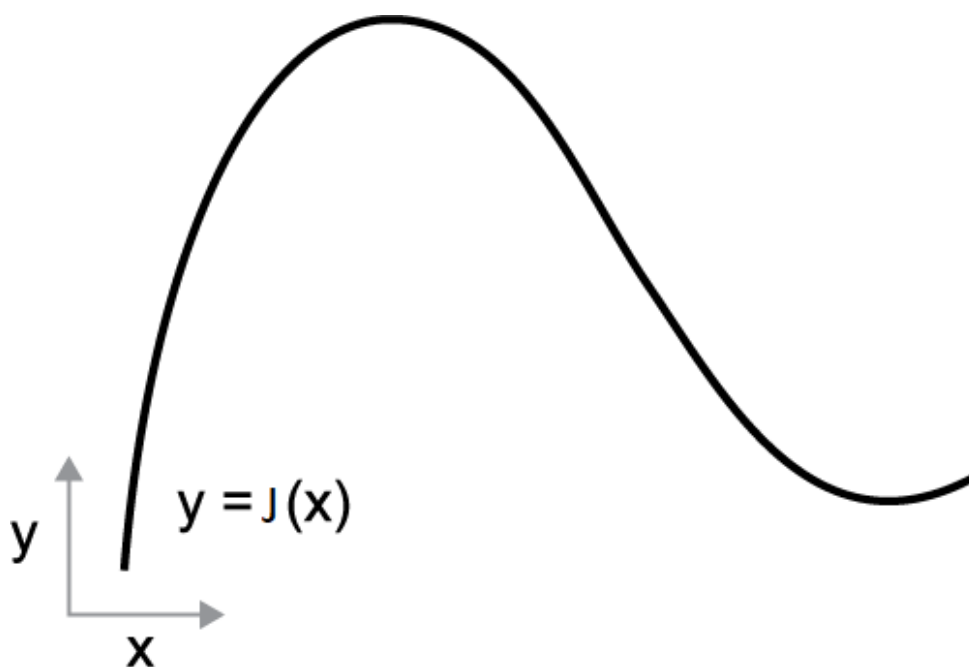
Gradients point uphill. We walk the other way.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

The figures follow Chollet, *Deep Learning with Python*, and Géron.

1. A derivative is a slope

Take a smooth map $J(x) = y$ that sends a number to a number. You can draw it as a curve in the plane.



Because J is **continuous**, a small change in x can only make a small change in y . Increase x by a small ε_x ; y moves by a small ε_y .

 A small step in x makes a small step in y .

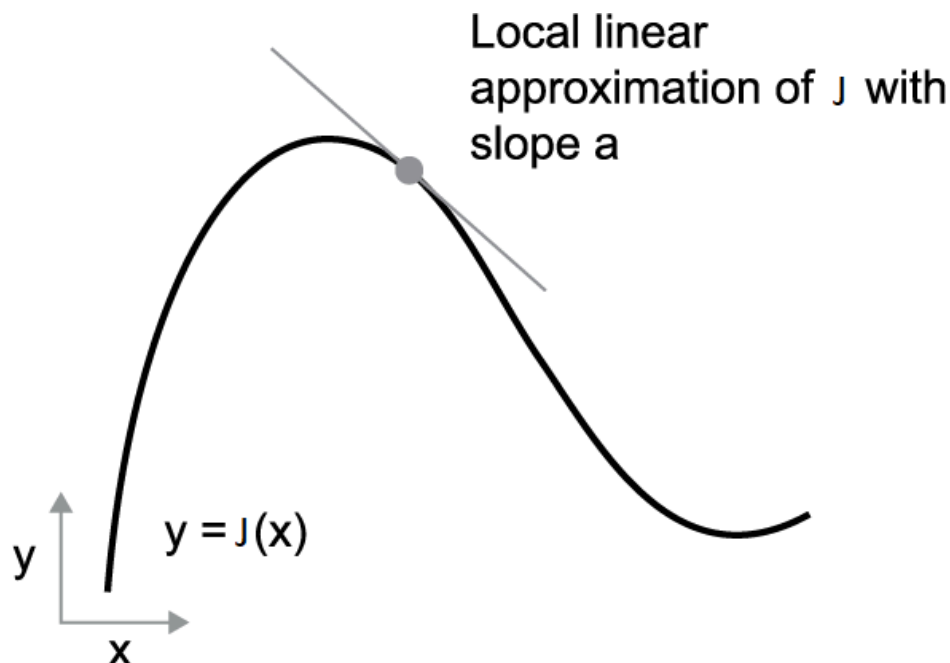
Because the curve has no kinks, when ε_x is small enough around a point p you may treat J as a line of slope a :

$$\varepsilon_y \approx a \varepsilon_x.$$

The number $a = J'(p)$ is the **derivative** at p .

- $a < 0$: a small increase in x **decreases** J .
- $a > 0$: a small increase in x **increases** J .

- $|a|$ (the **magnitude**) is how fast that happens.



To **minimize** J , walk against the sign of a . That one sentence is gradient descent in one dimension. Being able to differentiate is the main tool this course uses to find those x .

2. Continuity is not enough

Continuity says a small input change cannot jump the output. Smoothness says the graph has no corners, so a linear approximation exists. The algorithms in Module 1 assume at least that much. ReLU networks are allowed later; they are smooth almost everywhere. You still write a gradient.

3. Higher dimensions: the gradient

A map from one scalar to one scalar is a curve. A map $J(x, y) = z$ from a pair of scalars to a scalar is a **surface** in 3-space.

The analogue of a is the **gradient** ∇J : a vector of partial derivatives. It points to the direction of **steepest increase**. Descent uses $-\nabla J$.

Week 1 note **1.4** writes the gradient and the Hessian in the θ notation of the rest of the course. Week 2 uses both.

4. Practice

1. At a point where $J'(p) = 0$, what does a small step in x do to J , to first order? Why is that not enough to call p a minimum?

2. Why does a kink (a corner in the graph) break the “local line of slope α ” story?
3. If ∇J points uphill, what vector do you add to θ in a descent step, up to a step-size α ?